

Theory-Insight-Guided Empirical Research for Understanding Applied Combinatorial Solving

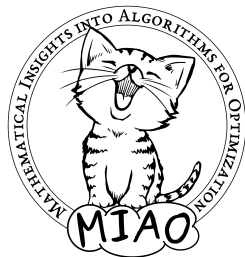
Jakob Nordström

University of Copenhagen and Lund University

Dagstuhl Seminar 26292

Analysis of Algorithms Beyond the Worst Case

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Me and My Research

Jakob Nordström

University of Copenhagen
and Lund University

`jn@di.ku.dk`

`www.jakobnordstrom.se`



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Research interests:

- Computational complexity theory
- Applied combinatorial optimization
- Certifying algorithms / proof logging



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Research interests:

- *Proving exponential lower bounds for NP-hard problems*
- *Designing algorithms to solve NP-hard problems in (ideally) linear time*
- *Generating machine-verifiable proofs that such algorithms reason correctly*



The Unreasonable Effectiveness of Combinatorial Solving

- Most combinatorial optimization problems are NP-hard
- The Boolean satisfiability (SAT) problems is widely believed to require exponential time [IP01, CIP09]
- However, dramatic improvements since the turn of the millennium has led to very efficient applied algorithms for
 - ▶ SAT solving [BHvMW21]
 - ▶ Constraint programming [RvBW06]
 - ▶ Mixed integer linear programming [AW13, BR07]
 - ▶ Satisfiability modulo theories (SMT) solving [BHvMW21]
- Can we use complexity theory to understand what is going on inside such combinatorial solvers?!

Theory and Practice Meeting Halfway?

Proof complexity

- Model reasoning used by algorithm as formal proof system
- Study efficient proofs for correct answers
- Non-existence of efficient proofs imply (strong) impossibility results, but can only be obtained for highly structured problems
- Existence of efficient proofs does not mean such proofs can be found algorithmically
- And proof systems can easily oversimplify algorithmic reasoning

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Theory-insight-guided empirical research

- Run computational experiments with solvers
- Check if performance matches predictions we cannot prove formally by using suitable problems from the proof complexity literature
- Analyze proofs generated from solver reasoning for “real-world” problems

TL;DR: *The workshop organizers made me do it. . .*

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Slightly longer version:

- This is a somewhat old line of research [JMNŽ12, EGG⁺18, EGNV18, KN20]
- Tons of hard work, but not an obviously rapturous reception. . .
- No proofs — just circumstantial evidence
- Results are open to interpretation/criticism
- But I personally still believe this type of research makes sense. . .
- So maybe it is worth talking about it, to see if this can at least generate some interesting discussions?

Contents of This Talk

Focus on Boolean satisfiability (SAT) solving

- ① Crash course on modern SAT solving
- ② Compare actual SAT solver performance on theoretical benchmark formulas with properties shown in proof complexity literature [EGG⁺18]
- ③ Study properties of actual proofs generated when solvers run on “real-world” benchmark formulas [KN20]

The Boolean Satisfiability (SAT) Problem

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge \\ (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

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- Variables should be set to **true** (= 1) or **false** (= 0)
- Constraints like $x \vee \bar{y} \vee z$ means x or z should be true or y false
- $\wedge$ means all constraints should hold simultaneously
- Is there a truth value assignment satisfying all constraints?

State-of-the-art SAT Solvers: Ingredients

Basic SAT solving method: backtracking search / DPLL [DP60, DLL62]

Many more ingredients in modern conflict-driven clause learning (CDCL) SAT solvers (as pioneered in [BS97, MS99, MMZ⁺01]), e.g.:

- Branching or decision heuristic (choice of pivot variables crucial)
- When backtracking, compute explanation for conflict and add to formula as new clause (clause learning)
- Every once in a while, restart from beginning (but save computed info)
- Preprocessing the formula before the search even starts

Variable Assignment Heuristics

Unit propagation

- Suppose current assignment ρ falsifies all literals in $C = \ell_1 \vee \ell_2 \vee \dots \vee \ell_k$ except one (say ℓ_k) — C is **unit under ρ**
- Then ℓ_k has to be true, so set it to true
- Known as **unit propagation** or **Boolean constraint propagation**
- Always propagate if possible — in modern solvers aim for $\approx 99\%$ of assignments being unit propagations

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VSIDS (Variable state independent decaying sum)

- When backtracking, score $+1$ for variables “causing conflict”
- Also multiply all scores with factor $\kappa < 1$ — exponential filter rewarding variables involved in recent conflicts
- When no propagations, **decide** on variable with highest score

Clause Learning

- At conflict, want to add clause avoiding same part of search tree being explored again
- Suppose we can compute that **decisions** $x = 1, y = 0, z = 1$ responsible for conflict
- Then can add $\bar{x} \vee y \vee \bar{z}$ to avoid these decisions being made again — **decision learning scheme** (this idea goes all the way back to [SS77])
- Nowadays, more sophisticated learning schemes starting with [MS99, MMZ⁺01]
- Often described in terms of cuts in **conflict graph**
- More helpful to view conflict analysis as **syntactic derivation** applied on clauses unit propagating to conflict

Decisions, Unit Propagations, and Conflict

Two kinds of assignments — illustrate on example formula:

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

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Free choice to assign value to variable

Notation $p \stackrel{d}{=} 0$

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Forced choice to avoid falsifying clause

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Continue until satisfying assignment or **conflict**

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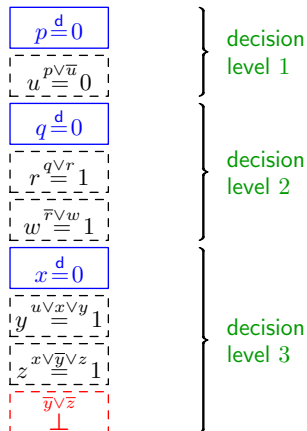
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Conflict Analysis

Time to analyse this conflict and learn from it!

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Could backtrack by erasing **conflict level** & flipping last decision (this is what DPLL does)

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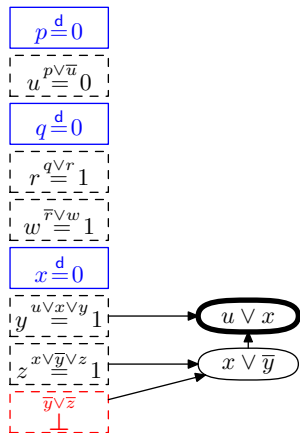
Case analysis for last two clauses over propagated variable:

- $x \vee \bar{y} \vee z$ wants $z = 1$
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- Merge clauses & remove z — must satisfy $x \vee \bar{y}$

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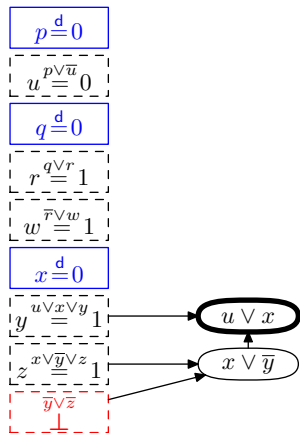
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Repeat until **UIP clause** with only 1 variable at conflict level after last decision — **learn** and **backjump**

Complete Example of CDCL Execution

Backjump: undo max #decisions while learned clause propagates

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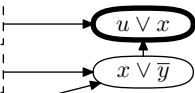
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$$\bar{y} \vee \bar{z} \stackrel{\perp}{=}$$



$$p \stackrel{d}{=} 0$$

$$u \stackrel{p \vee \bar{u}}{=} 0$$

Assertion level 1 (2nd largest level in learned clause) —
trim trail to that level

Complete Example of CDCL Execution

Backjump: undo max #decisions while learned clause propagates

$$(p \vee \bar{u}) \wedge (q \vee r) \wedge (\bar{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \bar{y} \vee z) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{p} \vee \bar{u})$$

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$$w \stackrel{\bar{r} \vee w}{=} 1$$

$$x \stackrel{d}{=} 0$$

$$y \stackrel{u \vee x \vee y}{=} 1$$

$$z \stackrel{x \vee \bar{y} \vee z}{=} 1$$

$$\bar{y} \vee \bar{z} \stackrel{}{=} \perp$$

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$$x \vee \bar{y}$$

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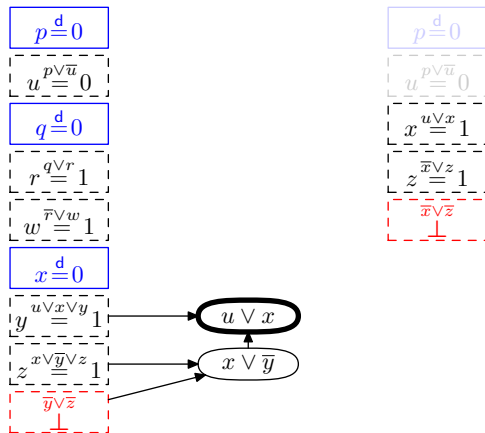
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Then continue as before. . .

Complete Example of CDCL Execution

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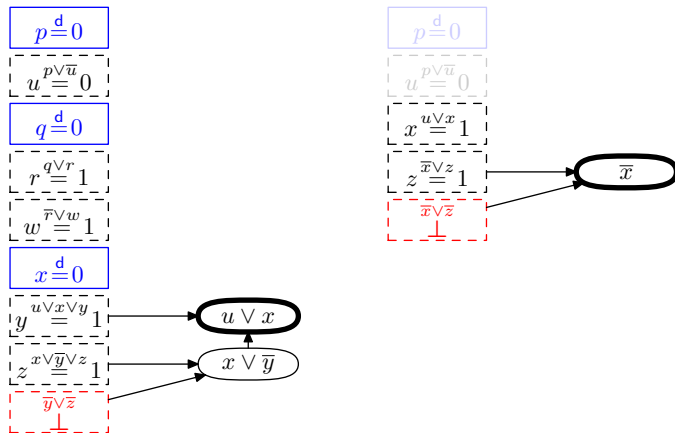
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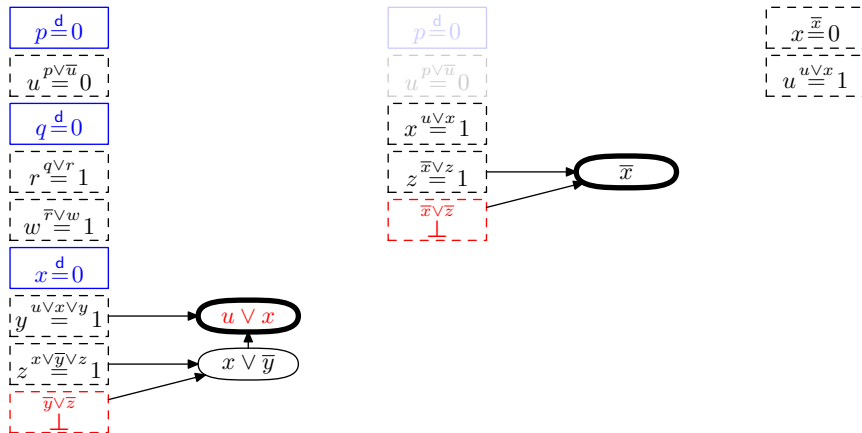
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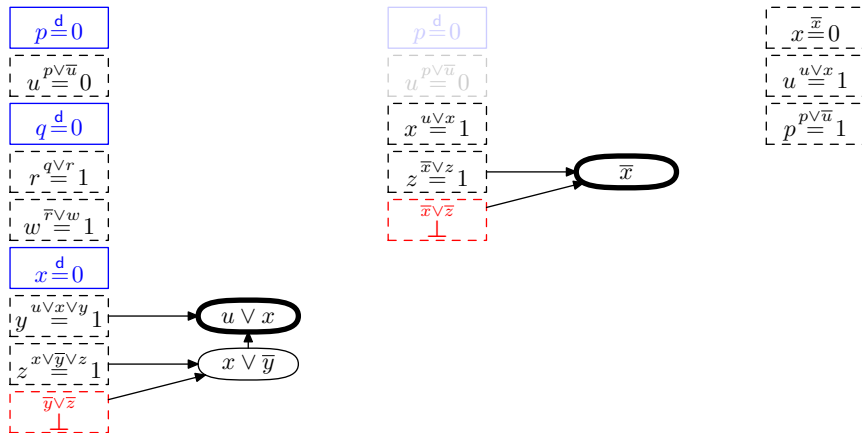
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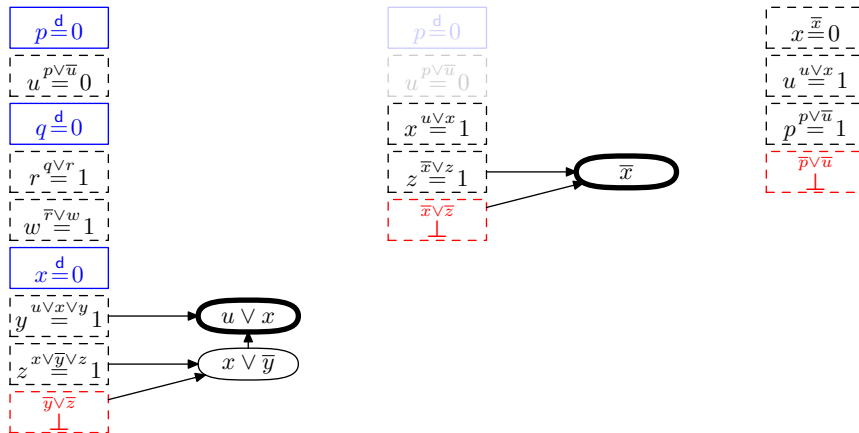
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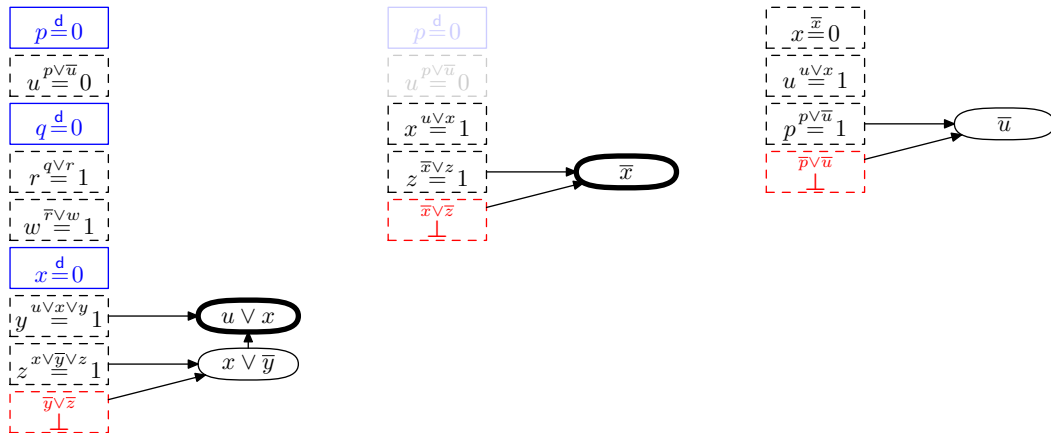
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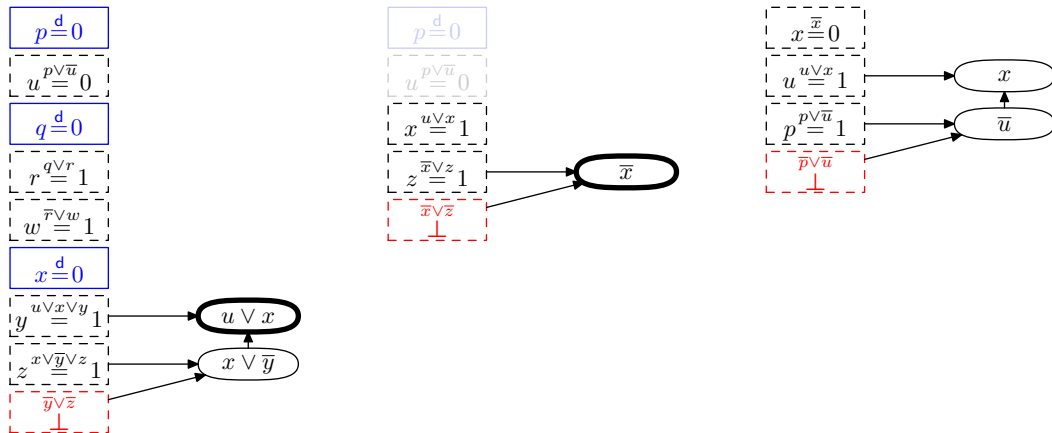
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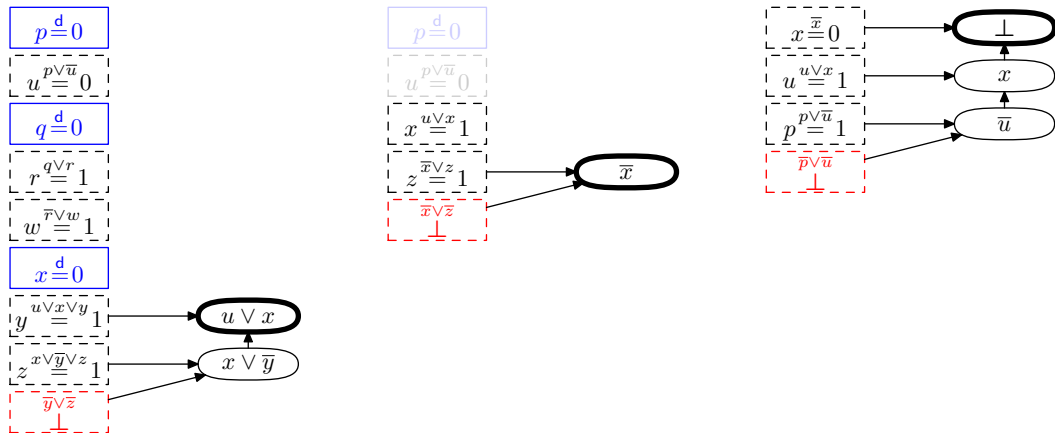
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Clause Database Reduction

- In addition to learning clauses, also **erase learned clauses that don't seem useful**
- Modern solvers do this **very aggressively**
- Speeds up CDCL search (in particular, unit propagation, which dominates running time)

Restarts

- Fairly frequently, **start search all over** (but keep learned clauses)
- Original intuition: stuck in bad part of search tree — go somewhere else
- Not the reason this is done now
- Popular variables with high VSIDS scores get set again [MMZ⁺01]
- Are even set to same values (**phase saving**) [PD07]
- Current intuition: improves the search by focusing on important variables
- Restart at fixed intervals or (better) make **adaptive restarts** depending on “quality” of learned clauses [AS09, AS12]

SAT Solver Analysis and the Resolution Proof System

How to make **rigorous** analysis of CDCL SAT solver performance?

Many intricate, hard-to-understand heuristics

So focus instead on **underlying method of reasoning**

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Resolution proof system [Bla37, Rob65]

- Start with clauses of CNF formula (**axioms**)
- Derive new clauses by **resolution rule**

$$\frac{C_1 \vee x \quad C_2 \vee \bar{x}}{C_1 \vee C_2}$$

Resolution Proofs by Contradiction

Resolution rule:

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Observation

If F is a satisfiable CNF formula and D is derived from clauses $D_1, D_2 \in F$ by the resolution rule, then $F \wedge D$ is satisfiable.

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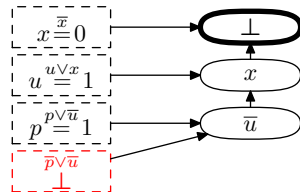
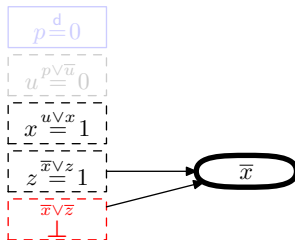
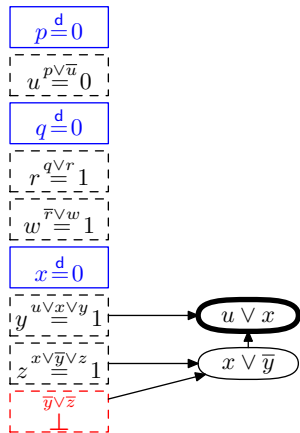
Such proof by contradiction also called **resolution refutation**

CDCL and Resolution Proofs

Obtain resolution proof. . .

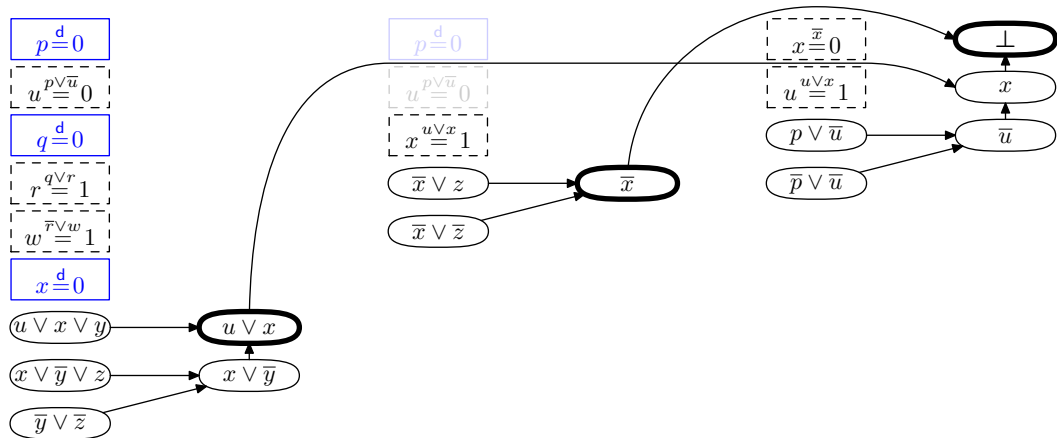
CDCL and Resolution Proofs

Obtain resolution proof from our example CDCL execution...



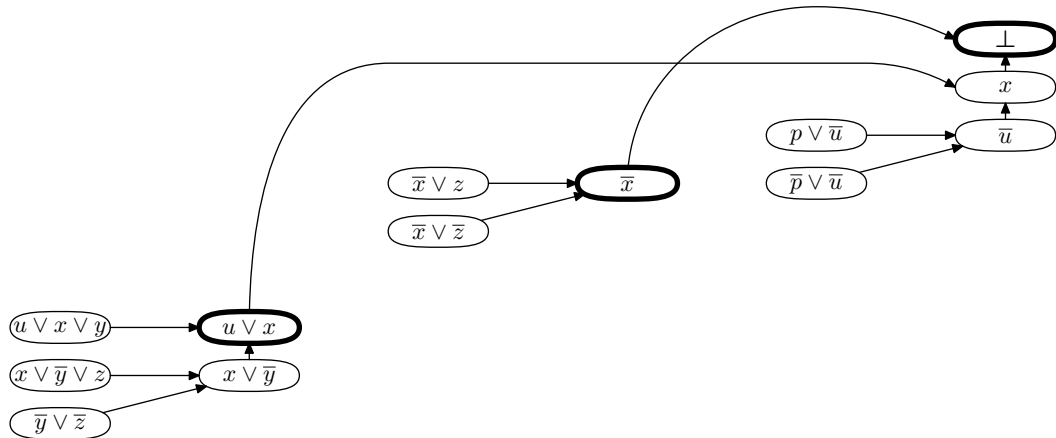
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(*) Except for some **preprocessing techniques**, which is an important omission, but this gets complicated and we don't have time to go into details. . .

CDCL and Efficient Proof Search

- Any CDCL solver reasoning (except preprocessing) can be captured by the resolution proof system
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(*) The details again get a bit too complicated for the purposes of this talk, but a key technical difficulty is to handle that solvers always greedily propagate

The Unreasonable Effectiveness of SAT Solvers

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- Very **poor theoretical understanding**:
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- Paradox: **resolution quite weak proof system**; many strong proof complexity lower bounds for (seemingly) “obvious” formulas encoding counting or parity arguments (e.g., [Hak85, Urq87, Pud97, BW01])

How to Analyse Performance of CDCL Solvers?

Applied approach

- **Instrument solver** with combinations of heuristics
- Run on **SAT competition benchmarks**
- Few and heterogeneous benchmarks
 - ▶ Poorly understood properties
 - ▶ Isolated data points
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Theoretical approach

- Relate CDCL performance to complexity measures for **resolution proof system?**
- Only considers existence of proofs, not algorithmic search
- Only asymptotic results (sometimes for gigantic formulas)

Why Not Combine Theory and Practice?

Our combined approach using theoretical benchmark formulas [EGG⁺18]

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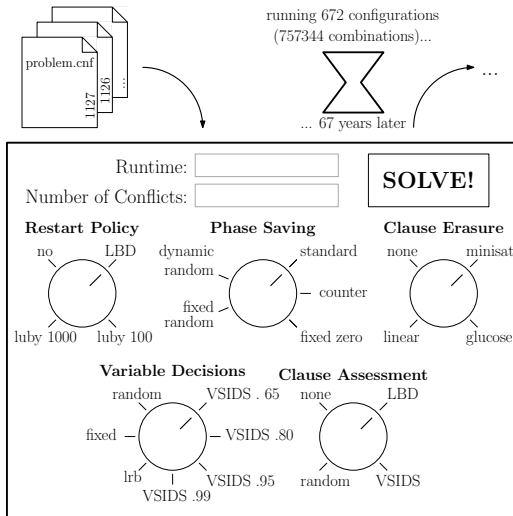
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⇒ Different benchmarks will **“stress-test” different heuristics**
- Code up **instrumented solver** with wide selection of heuristics
 - ▶ More complicated than it sounds — some heuristics tightly integrated
 - ▶ Run on (essentially full) cross product of heuristics
 - ▶ Which heuristics are important when?
 - ▶ How do heuristics interact?

Some Possible Objections

- How do we know that we are observing proof-theoretic properties as opposed to other structural characteristics of the formulas?
- Especially since sometimes benchmarks have to be tweaked so that theoretical guarantees are lost
- Theoretical formulas are not relevant — SAT solvers are designed to perform well on “real-world” benchmarks
- All benchmark formulas are unsatisfiable, but performance on satisfiable formulas is also important in practice
- And correlation is not causation

Our Set-up



... And the Results...

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- 1127 instances in 27 formula families
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This talk: Focus on some example findings

- Clause learning
- Restarts
- Memory management
- Branching heuristic

Theory: Clause Learning and Tree-Like Resolution

- Search in CDCL solvers **crucially guided by conflicts**
- Clause learning influences
 - ▶ variable decisions
 - ▶ restart policy
 - ▶ memory management
 - ▶ and more...
- Is **storing clauses learned** during conflict analysis also essential?

What theory says

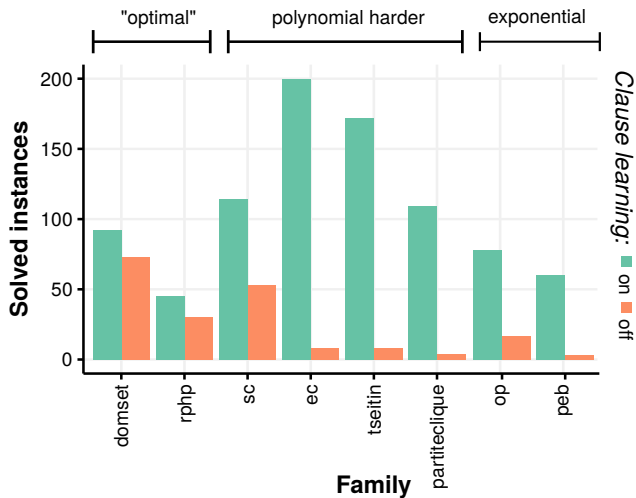
- no learning $\Rightarrow$ tree-like resolution (DPLL)

How to check in practice?

Run experiments on

- instances that separate tree-like and general resolution
- instances where tree-like is (asymptotically) optimal

Empirical: Clause Learning and Tree-Like Resolution



Restarts

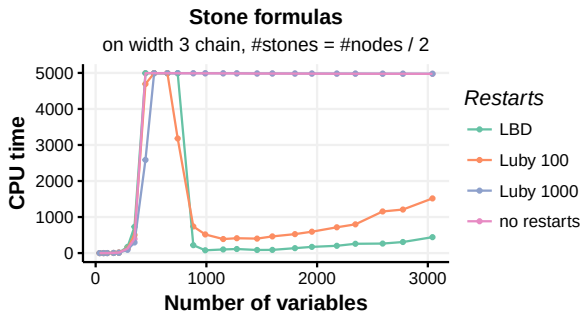
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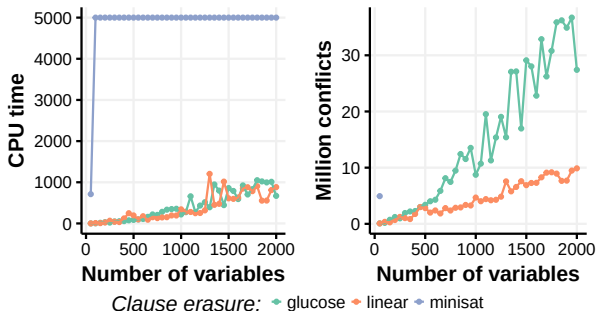
Memory Management: Time-Space Trade-offs

- CDCL solvers very aggressively minimize memory usage
- Dramatic time-space trade-offs in theory [BN11, BBI12, BNT13]
- Could such trade-off phenomena show up in practice? Seems like yes

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- CDCL solvers very aggressively minimize memory usage
- Dramatic time-space trade-offs in theory [BN11, BBI12, BNT13]
- Could such trade-off phenomena show up in practice? Seems like yes

Tseitin formulas on grid graphs (5 rows)



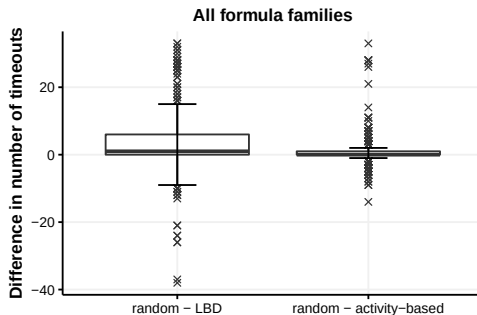
Database size: minisat ($\sim N^{0.25}$) < glucose ($\sim N^{0.5}$) < linear ($\sim N$) [$N = \# \text{conflicts}$]

Memory Management: Clause Assessment

- When time to purge database, how to **identify useful clauses** to keep?
 - ▶ Activity-based [ES04]
 - ▶ Literal block distance (LBD) [AS09]
 - ▶ Random (control)

Memory Management: Clause Assessment

- When time to purge database, how to **identify useful clauses** to keep?
 - ▶ Activity-based [ES04]
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 - ▶ Random (control)
- LBD quite successful, especially when space is getting tight
- Activity-based indistinguishable from random (& random OK-ish)

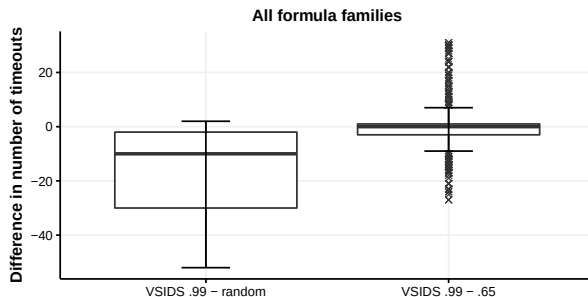


Branching Heuristics: Importance of Memory Horizon

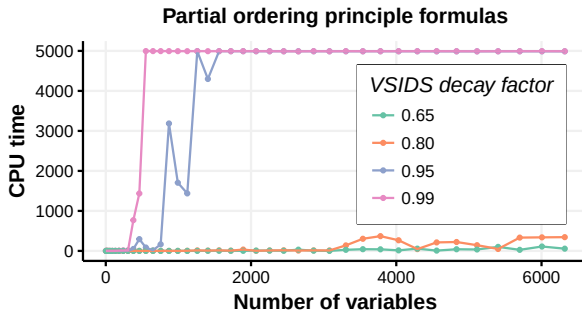
- Branch on variable appearing **most often in recent conflicts**
 - + **exponential decay** to reward recent conflicts: VSIDS [MMZ⁺01]
 - ▶ Low VSIDS factor = short memory
 - ▶ High VSIDS factor = long memory
- Choice high/low depends on type of formulas
- But random decisions are pretty much always bad

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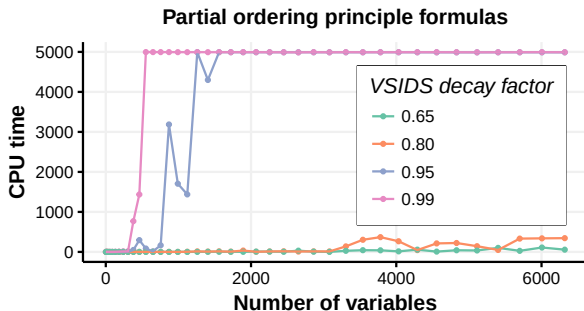
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Branching Heuristics: A Particularly Dramatic Example



Branching Heuristics: A Particularly Dramatic Example



- What's going on? We're not sure...
 - Instances very hard for tree-like resolution ($\approx$ DPLL) [BG01, BW01]
 - Low VSIDS factor $\Rightarrow$ search focus locally on latest conflicts
 - Maybe high VSIDS factor $\Rightarrow$ closer to DPLL-style search?!
- (Consistent with our experiments, but more data & insights needed)

Second Approach: Analyze Solver Reasoning for Applied Benchmarks

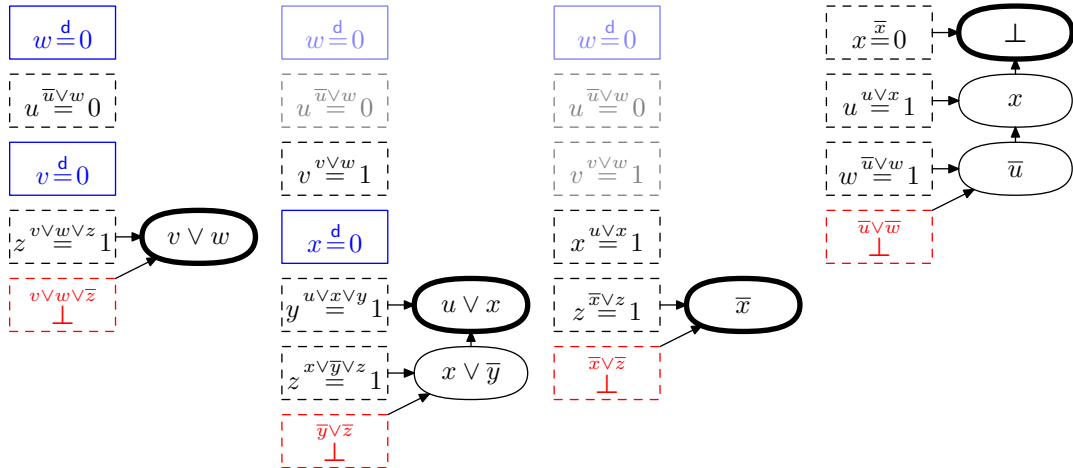
Study solver-generated proofs [KN20]

Measure impact of CDCL solver heuristics by

- Running SAT solvers with different settings on real-world benchmarks
- Looking at resolution proofs produced by solvers
- Smaller proof $\Rightarrow$ more efficient reasoning
- Also trim proofs to determine which clauses needed to reach final conclusion

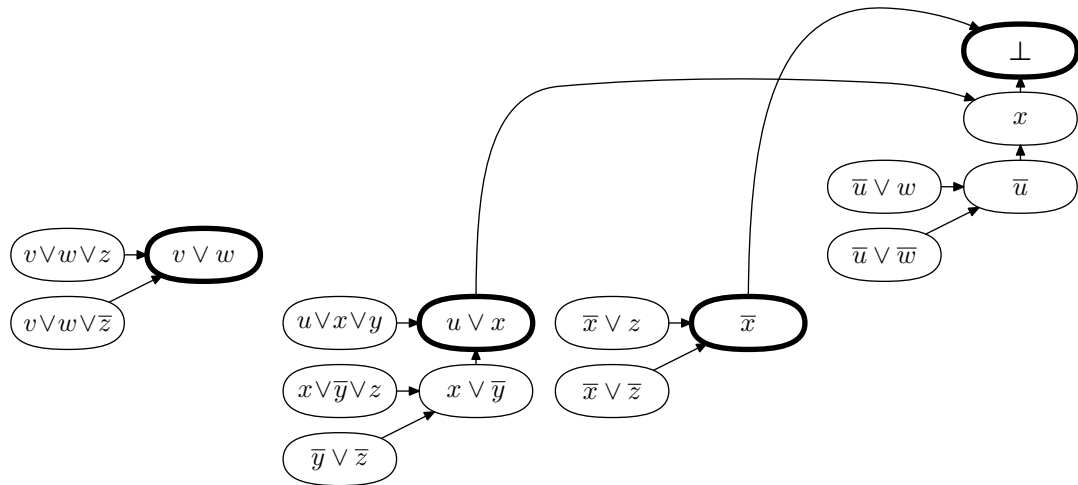
Proof Trimming by Picture

$$(u \vee x \vee y) \wedge (v \vee w \vee z) \wedge (x \vee \bar{y} \vee z) \wedge (v \vee w \vee \bar{z}) \wedge (\bar{x} \vee z) \wedge (\bar{y} \vee \bar{z}) \wedge (\bar{x} \vee \bar{z}) \wedge (\bar{u} \vee w) \wedge (\bar{u} \vee \bar{w})$$



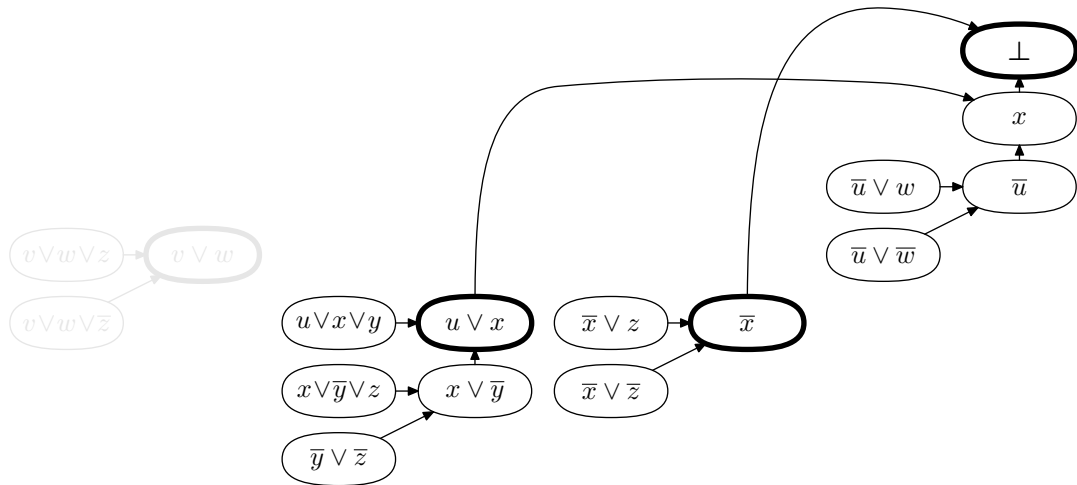
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Heuristics Studied

Restarts: Adaptive or non-adaptive

- Adaptive based on literal block distance (LBD)
- Fixed Luby restarts — multiples of sequence $(1, 1, 2, 1, 1, 2, 4, 1, 1, 2, 1, 1, 2, 4, 8, \dots)$

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Clause erasure: On/off

- Standard clause database reduction policy
- No clause erasures

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Clause assessment: Correlate usefulness of clauses with

- Literal block distance (LBD) score
- Conflict activity
- Clause size
- Decision level learned
- Backjump length caused
- # conflicts since last restart

Focus on clauses of size ≥ 3 , since smaller clauses will never be removed

Benchmarks Selection

- Randomly sample 200 benchmarks from recent SAT competitions
- Then filter so that all relevant solver configurations can solve all benchmarks
- Concretely, for each experiment:
 - ▶ Let all configurations pick their 150 best benchmarks
 - ▶ Take intersection for all solver configurations used in experiment
 - ▶ Use whatever time it takes to get all solvers solve all these benchmarks
- In the end, 120–133 benchmarks per experiment
- And again fairly massive computations. . .

Non-adaptive restarts (Luby sequence)

- Higher restart frequency makes both untrimmed and trimmed proofs smaller

Analyzing Restarts Using Solver-Generated Proofs

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- Not so significant size decrease for trimmed proofs
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General conclusion

- Restarts improve reasoning power of solvers

Effects of switching off clause erasure

- Often smaller untrimmed proofs (but not always!)
- No significant effect on trimmed proofs
- So core reasoning not stronger with more clauses in memory, but extra clauses help solver focus and avoid work that turns out to be useless with hindsight(?)

Which clauses are likely to contribute in proofs?

- Very low (good) LBD score excellent predictor
- Otherwise conflict activity is good indicator
- Random choice is clearly worse than both (difference from theory benchmarks)
- Other features measured did not have strong impact
- Lends more rigorous support for currently most popular clause database management heuristic [Oh16]

Clause Assessment in More Detail: Two types of experiments

① With clause deletions

- ▶ Fix non-adaptive strategy for # clauses to remove at database reduction time
- ▶ Decide which clauses to erase based on LBD, conflict activity, or randomly
- ▶ Findings:
 - ★ Proof size is smaller for LBD-based erasure than activity-based erasure
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② Without clause deletions

- ▶ Study untrimmed and trimmed proofs
- ▶ Clauses are “useful” if they remain in trimmed proof
- ▶ Similar ideas used for analysis in [Sim14, MY16]
- ▶ Findings:
 - ★ Very good LBD score correlates with surviving trimming
 - ★ Conflict activity better predictor over broader range of values
 - ★ LBD better measure than simply clause size
- ▶ Agrees with clause database management heuristics proposed in [Oh16]

Turn circumstantial evidence into hard mathematical proofs

- Determine whether restarts increase the theoretical reasoning power of SAT solvers
- Finds ways of formalizing other findings

Scope for Theoretical Research

Turn circumstantial evidence into hard mathematical proofs

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Explain why SAT is “easy in practice”

- Is it possible to give mathematical characterization of properties of “real-world problems”?
- Or to find interesting synthetic benchmarks that look like applied problems?
- Can one prove that CDCL SAT solvers run fast for interesting classes of formulas?

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Find proof systems that tightly characterize other solving paradigms

- MaxSAT and pseudo-Boolean solving
- 0–1 integer linear and mixed integer linear programming

Take-Away Message

This work

- Goal: Not to improve SAT solvers, but to **understand** how the best solvers work
- SAT solvers are too hard to model, so run **computational experiments**
- Use **crafted benchmark formulas** to check if performance matches theory predictions
- Or analyze **solver-generated proofs** for “real-world” formulas

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Some possible directions for future research

- Analyze proofs for applied CNF formulas in more detail
- Extract and analyze “proofs” for satisfiable formulas(?)
- MaxSAT and pseudo-Boolean solvers
 - ▶ Study performance on crafted benchmarks
 - ▶ Analyze proofs for “real-world” formulas for
 - ▶ Use results to improve heuristics
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Thank you for your attention!

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